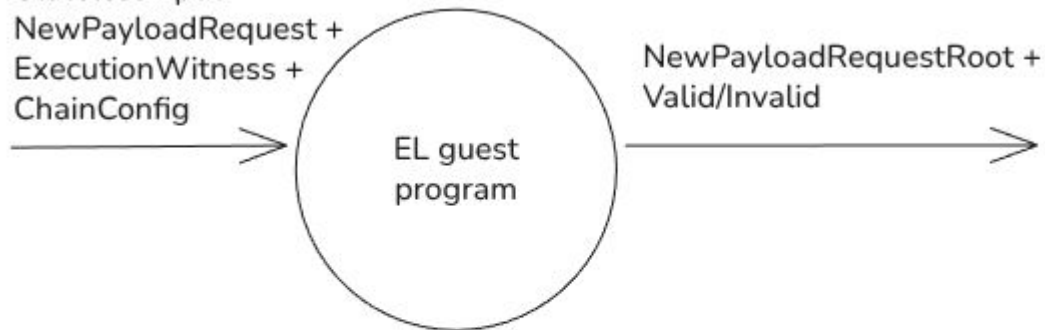


zkEVM project 1

Execution Witness

Project TL;DR

StatelessInput =
NewPayloadRequest +
ExecutionWitness +
ChainConfig



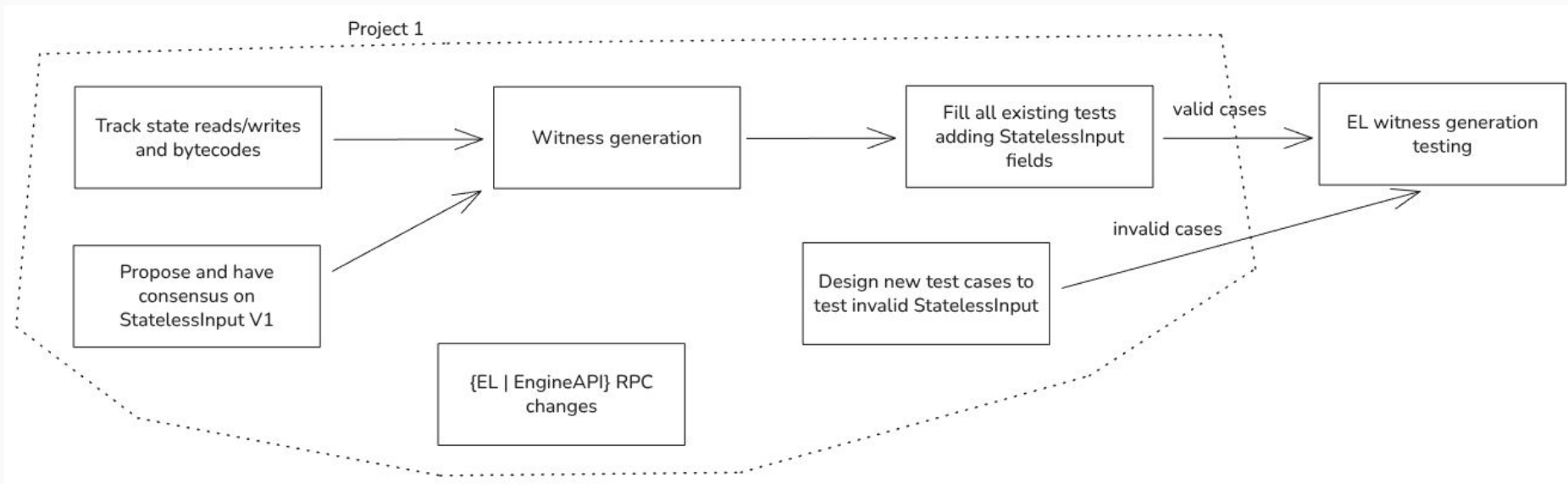
ExecutionWitness:

- ancestor headers
- pre-state multi-proofs
- post-state aux nodes
- bytecodes

Goals:

- Standardized ExecutionWitness
- RPC compatibility & latency
- Tests

Roadmap-ish?



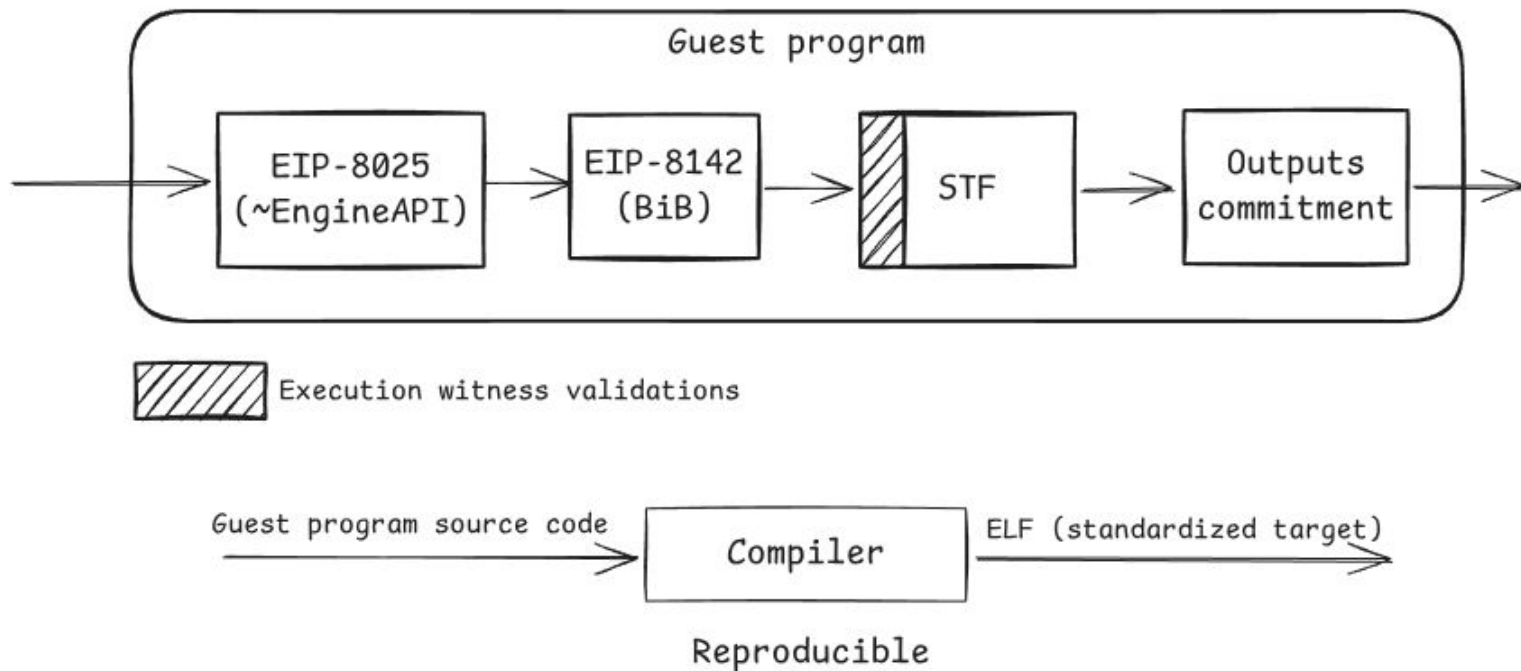
Updates

- Specs:
 - Peter: [generalized state tracker PR](#)
 - Ignacio: [add ancestors and bytecode to Peter's state tracker tracker PR](#)
 - Ignacio: [full e2e witness generation and integration in EELS/EEST PoC](#)
- Research:
 - Ignacio: [deep dive in OpenVM alternative `nodes` serialization and MPT optimizations](#)

zkEVM project 2

Guest Program

Project TL;DR



Updates

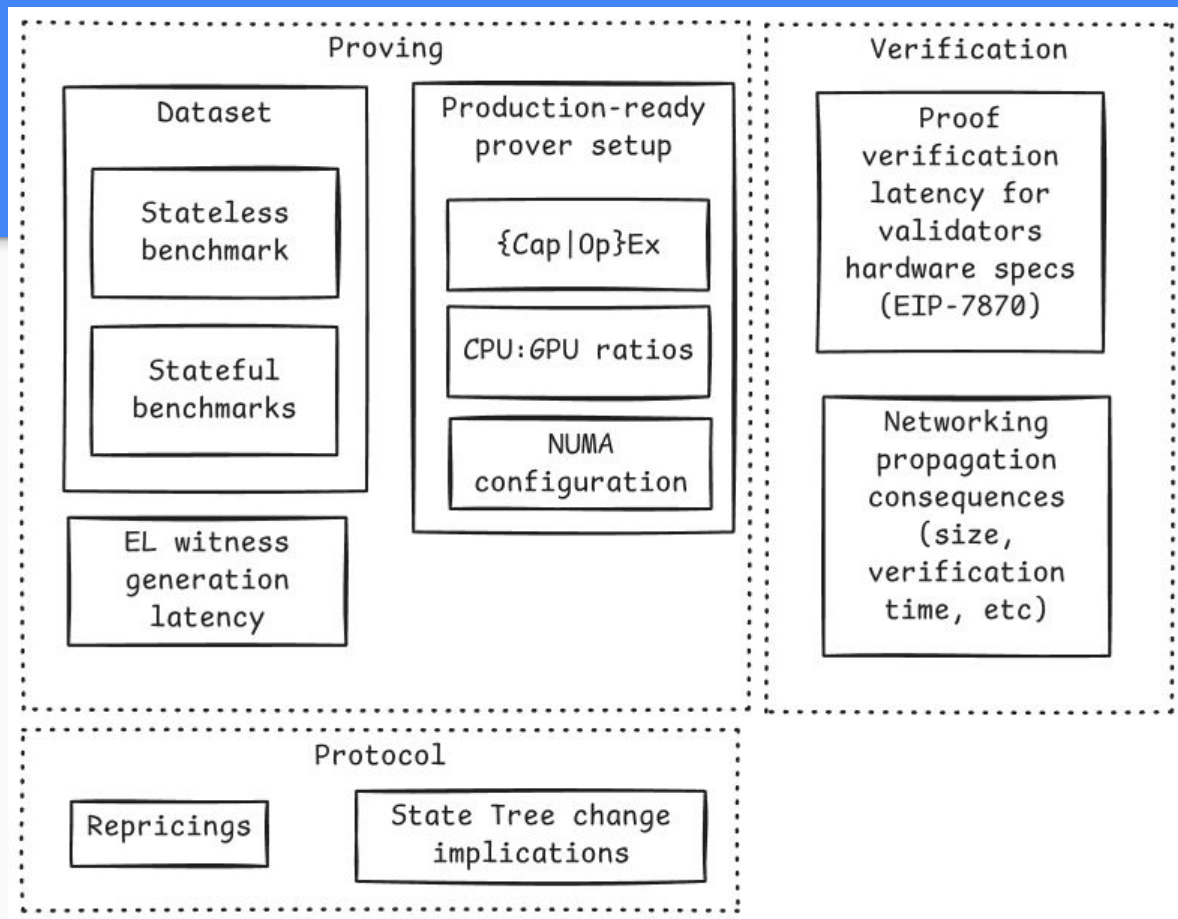
- Guest program in execution specs:
 - Mario/Peter: [execution-spec refactoring requirements](#) and [meta-issue](#)
- Reproducible ELF builds and distribution:
 - DeveloperUche: An [automated test validating ELF reproducibility](#).
 - DeveloperUche: [Minisig integration](#) in ere-guests & [binary for ELF verification](#).
- Clients updates:
 - Ethrex / Ivan: [ELF distribution pipeline](#) & [Guest program refactor](#)
 - {Ethrex & Reth} Ignacio: [Implement EIP-8025](#)

zkEVM project 6

Benchmarking and Metrics



Project TL;DR



Updates

- Benchmark guest programs against zkVMs locally:
 - Ignacio: [Zisk GPU scaling exploration](#)
 - Stefan & Han: NUMA setup for Zisk which can explain bottlenecks in the document above.
 - Stefan: Options on how to buy hardware for production-ready provers.
 - EEST: benchmarks now [support Osaka](#).
- Investigate multidimensional metering/pricing:
 - Ignacio: Protocol repricings series ([Part 1](#) and [Part 2](#))